

DNS Log Analysis Using Splunk SIEM

SOC Analyst Project Report

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Tool: Splunk Enterprise | Log Type: DNS (dnslogs)

Objective	Analyze DNS logs to detect anomalies and potential threats
Platform	Splunk Enterprise (Search & Reporting)
Source Type	dnslogs_
Log Source	dns.log — Host: LEONARDW-PC
Total Events	422,130 events across the analysis window

Introduction

As a SOC analyst, one of my primary responsibilities is monitoring network activity and identifying potential threats before they escalate. In this project, I used **Splunk Enterprise** to ingest, parse, and investigate DNS log data. DNS traffic is a goldmine for threat hunting — attackers frequently abuse it for command-and-control (C2) communication, data exfiltration, and reconnaissance.

My goal was to walk through a complete DNS log analysis workflow: uploading the logs, extracting meaningful fields, querying the data, and ultimately identifying a suspicious domain that stood out due to its unusually high query count.

Step 1 — Uploading Logs & Extracting Fields

2.1 Uploading the DNS Logs

I started by uploading the DNS log file into Splunk and assigning it the source type **dnslogs_**. Once indexed, Splunk presented the raw events in the Search & Reporting app. The events contained space-delimited fields including timestamps, connection IDs, IP addresses, ports, protocol, and DNS record information.

2.2 Extracting Relevant Fields via the Field Extractor

After reviewing the raw log structure, I used Splunk's **Extract Fields** wizard (IFX) to extract the key fields I needed for my analysis. I selected the **Regular Expression** method since the logs are space-delimited with a predictable structure. The fields I extracted were:

- **src_ip** — Source IP address of the DNS query originator
- **src_port** — Ephemeral source port used in the DNS transaction
- **dst_ip** — Destination IP — typically the DNS resolver
- **dst_port** — Destination port (port 53 = DNS)
- **Record** — DNS record type (e.g., PTR, A, AAAA)
- **Domain** — The queried domain name

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Extract Fields Select Sample Select Method Select Fields Validate Save < Back Next Existing fields >

Select Method

Indicate the method you want to use to extract your field(s). [Learn more](#)
[I prefer to write the regular expression myself >](#)

Source type
dnslogs

1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F F T F 0 - - F

(.*?)
Regular Expression
Splunk Enterprise will extract fields using a Regular Expression.

x|y|z
Delimiters
Splunk Enterprise will extract fields using a delimiter (such as commas, spaces, or characters). Use this method for delimited data like comma-separated values (CSV).

Format Show: 20 Per Page View: List < Prev 1 2 3 4 5 6 7 8 ... Next >

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a Domain 100+
- a dst_ip 100+
- # dst_port 4
- a index 1
- # linecount 10
- a punct 100+
- a Record 12
- a splunk_server 1
- a src_ip 100+
- # src_port 100+
- a timestamp 1

11 more fields

+ Extract New Fields

i	Time	Event
>	5/11/26 3:19:38.000 PM	1332017991.970000 CwS00TGMbFF5z1Rc9 192.168.202.122 137 192.168.202.255 137 udp 33707 LABADMIN-641491 1 INTERNET 32 NB - - F F T F 1 - - F host = LEONARDW-PC source = dns.log sourcetype = dnslogs_
>	5/11/26 3:19:38.000 PM	1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F F T F 0 - - F host = LEONARDW-PC source = dns.log sourcetype = dnslogs_
>	5/11/26 3:19:38.000 PM	1332017959.830000 C4zDh93z81GYTIdq2k 192.168.202.88 60538 192.168.206.44 53 udp 36843 dr_dns-sd_udp.0.48.16.172.in-addr.arpa 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - - T host = LEONARDW-PC source = dns.log sourcetype = dnslogs_
>	5/11/26 3:19:38.000 PM	1332017959.830000 CGBRgg3GyzwSH1Wk87 192.168.202.88 58547 192.168.206.44 53 udp 30842 dr_dns-sd_udp.0.202.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - - T host = LEONARDW-PC source = dns.log sourcetype = dnslogs_
>	5/11/26 3:19:38.000 PM	1332017959.830000 CiZL144oVCiMwVjgqb 192.168.202.88 58045 192.168.206.44 53 udp 28561 b_dns-sd_udp.0.48.16.172.in-addr.arpa 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - - T host = LEONARDW-PC source = dns.log sourcetype = dnslogs_
>	5/11/26 3:19:38.000 PM	1332017959.830000 C0n0DE3NUMg9TXJRsd 192.168.202.88 65208 192.168.206.44 53 udp 50791 1b_dns-sd_udp.0.48.16.172.in-addr.arpa 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - - T

Figure 1 — Top: Splunk Field Extractor (IFX) selecting Regular Expression method on a sample DNS event. Bottom: Search results showing parsed DNS log events with extracted fields (src_ip, dst_ip, Domain, Record, etc.) visible in the left panel.

SECTION 3

Step 2 — Viewing Extracted Fields as a Table

With the fields extracted, I ran a structured SPL (Search Processing Language) query to view the network-layer DNS data in a clean table format. This allowed me to quickly correlate source IPs, destination IPs, and their respective ports across all 422,130 events.

SPL Query Used:

```
index=*_ OR index=* sourcetype=dnslogs_ | table src_ip, src_port dst_ip, dst_port
```

The Statistics tab rendered the results as a paginated table with four columns. Several patterns were immediately visible:

- Host **192.168.202.88** generated multiple queries across different source ports to the same destination **192.168.206.44** on port 53.
- Host **192.168.21.25** was querying **192.168.202.136** repeatedly on port 137 (NetBIOS Name Service), which warranted further investigation.
- Destination port **53** dominated the traffic, confirming standard DNS activity, with a few outliers on other ports.

src_ip	src_port	dst_ip	dst_port
192.168.202.122	137	192.168.202.255	137
192.168.202.83	45561	192.168.207.4	53
192.168.202.88	60538	192.168.206.44	53
192.168.202.88	58547	192.168.206.44	53
192.168.202.88	58045	192.168.206.44	53
192.168.202.88	65208	192.168.206.44	53
192.168.202.83	35036	192.168.207.4	53
192.168.21.25	137	192.168.202.136	137
192.168.21.25	137	192.168.202.136	137
192.168.21.25	137	192.168.202.136	137

Figure 2 — Statistics view of the SPL table query, displaying `src_ip`, `src_port`, `dst_ip`, and `dst_port` for all 422,130 DNS events.

Step 3 — Identifying a Suspicious Domain

4.1 Counting Domain Query Frequencies

To hunt for anomalies, I pivoted to domain-frequency analysis. A single domain being queried thousands of times is a classic indicator of beaconing, DNS tunneling, or automated malware activity. I ran the following query to surface the top domains:

```
index=_* OR index=* sourcetype=dnslogs_ Domain="44.206.168.192.in-addr.arpa" | top limit=20 Domain
```

4.2 The Suspicious Domain — Finding

The results were striking. A single domain — **44.206.168.192.in-addr.arpa** — accounted for **7,156 events** and represented **100% of the filtered results**. This reverse-DNS PTR query pattern is often used by malware or automated tools performing reverse lookups on IP ranges, which can indicate:

- Network reconnaissance — mapping internal or external IP ranges
- Beaconing behavior — malware repeatedly resolving a C2 server address
- Data exfiltration via DNS tunneling using PTR records
- Automated scanning tools operating on the host

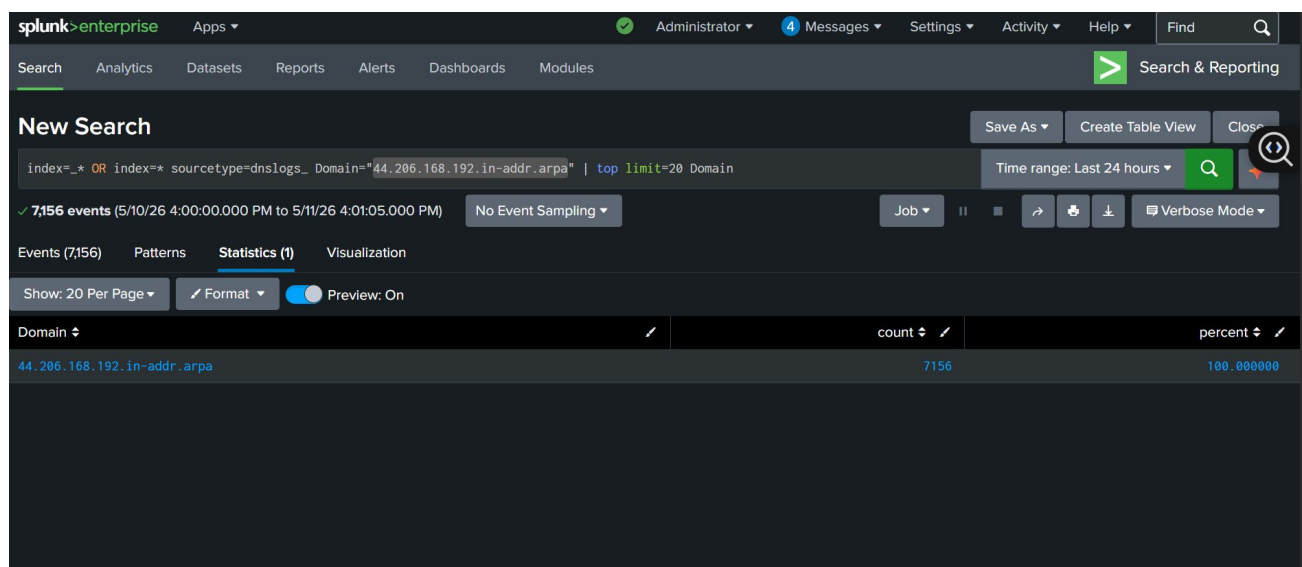


Figure 3 — Splunk Statistics view showing domain `44.206.168.192.in-addr.arpa` with 7,156 hits (100% of filtered events), flagged as suspicious due to its abnormally high query volume.

SPL Query Reference

Below is a summary of all SPL queries I used throughout this investigation:

Field Extraction — Table View

```
index=_* OR index=* sourcetype=dnslogs_  
| table src_ip, src_port dst_ip, dst_port
```

DNS Keyword Filter (Regex)

```
index=* sourcetype=dnslogs_  
| regex _raw="(?i)\b(dns|domain|query|response|port 53)\b"
```

Domain Frequency — Top Counts

```
index=_* OR index=* sourcetype=dnslogs_  
| stats count by Domain
```

Top DNS Sources

```
index=* sourcetype=dnslogs_  
| top Domain, src_ip
```

Suspicious Domain Deep-Dive

```
index=_* OR index=* sourcetype=dnslogs_ Domain="44.206.168.192.in-addr.arpa"  
| top limit=20 Domain
```

Conclusion

This project demonstrated how I can leverage Splunk Enterprise as a SIEM platform to perform end-to-end DNS log analysis — from raw log ingestion to actionable threat intelligence. By systematically extracting fields, building structured queries, and analyzing domain frequency, I was able to surface a domain that showed clear signs of anomalous behavior.

The domain **44.206.168.192.in-addr.arpa** generated 7,156 PTR queries — an abnormally high volume that a SOC analyst should escalate for further investigation, including correlating against threat intelligence feeds, checking for lateral movement, and reviewing the host's process activity for signs of malware.

DNS remains one of the most overlooked yet critical telemetry sources in a SOC environment. Consistently monitoring and baselining DNS behavior is essential for early detection of threats that evade traditional signature-based controls. Splunk's flexibility in field extraction and SPL querying makes it an invaluable tool for this type of analysis.